



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE 'A'

Course Code & Title: CS8392 Object Oriented Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Input and Output- Reading and Writing Files**

- **Course Outcome: CO4**

- **Topic Learning Outcome: TLO8**

- **Activity Chosen: Crossword Puzzle**

- **Justification:**

In Java different types of input and output stream are available. In order to remember the different classes used for input and output stream and file processing, crossword puzzle is used instead of normal way of summary.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- A crossword is an interesting word puzzle activity which helps the students to recollect the concept and try to find out the answer individually. It is one type of game, so this an enjoyable activity rather than regular multiple choice questions test.
- The activity was conducted in a classroom for a duration of 10 minutes.
- The students were given with a crossword puzzle of 10 questions in a paper. The questions were from the Input and output basics. They have to think and write the answers in the given puzzle.
- The activity was conducted during the last 10 minutes of the class.
- After the completion the answers were corrected and given to the students. Sample answers were shown in Figure:1-3.
- The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO4	3	3	2	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative	PO1	PO2	PO3	PO9	PSO1
------------	-----	-----	-----	-----	------

practice	3	3	2	2	1
Justification for correlation	The students apply the knowledge of Files	The students evaluate given problem statements and identify solutions	The students find the solutions for the given task.	The students demonstrate problem-solving skills as a team.	The students use the Files concepts to store and retrieve data to solve problems

• **Images / Screenshot of the practice:**

Crossword Puzzle- Input Output

Across

7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionally for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

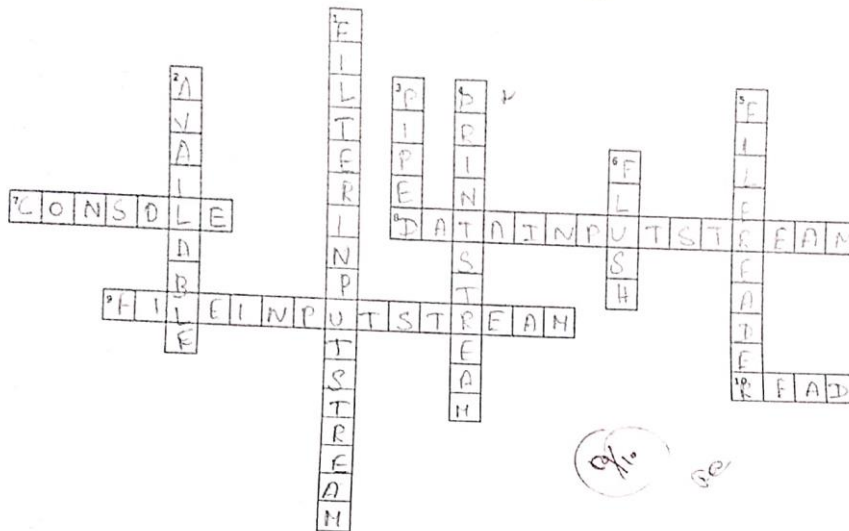
By: 953619106016
953619106015

Fig 1: Sample crossword puzzle

Crossword Puzzle- Input Output

953619106001

953619106026



Across

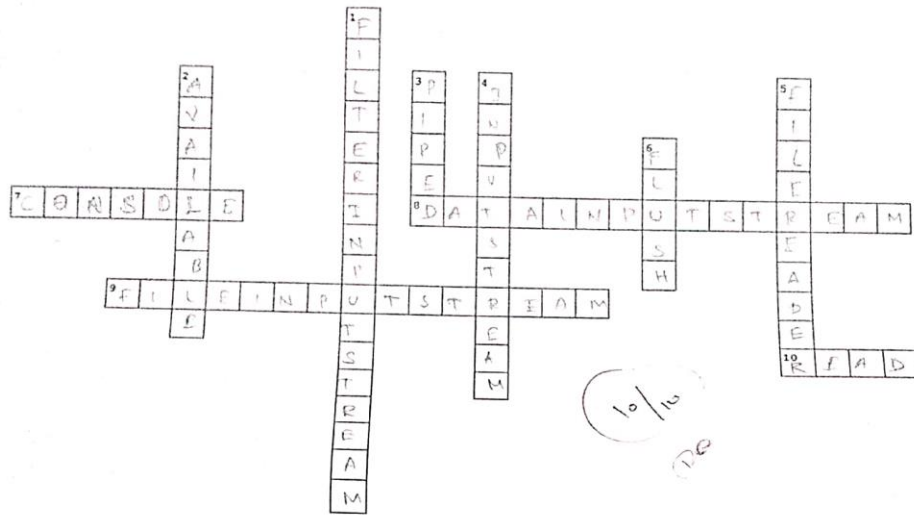
7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionality for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

Fig 2: Sample crossword puzzle

Crossword Puzzle- Input Output



Across

7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionality for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

Fig 3: Sample crossword puzzle

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Some of the students find very easy to identify the classes used for input and output.
- Some of them take time to complete the puzzle need more clarification on the various classes.

❖ Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The student level of understanding on various input output stream classes improved. They can easily identify the place where they used which type of stream classes.

❖ Challenges faced in implementation:

- Some students didn't understand the two type of IO stream classes struggle to find answer to the puzzle.
- Some of them feel difficult to identify the file classes and its method such as reading and writing.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/48.GE8151_Crossword.pdf



Department of Computer Science and Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE 'B'

Course Code & Title: CS8392 Object Oriented Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Input and Output- Reading and Writing Files**

- **Course Outcome: CO4**

- **Topic Learning Outcome: TLO8**

- **Activity Chosen: Crossword Puzzle**

- **Justification:**

In Java different types of input and output stream are available. In order to remember the different classes used for input and output stream and file processing, crossword puzzle is used instead of normal way of summary.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- A crossword is an interesting word puzzle activity which helps the students to recollect the concept and try to find out the answer individually. It is one type of game, so this an enjoyable activity rather than regular multiple choice questions test.
- The activity was conducted in a classroom for a duration of 10 minutes. The students were given with a crossword puzzle of 10 questions in a paper. They have to think and write the answers in the given puzzle.
-
- The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in figure 2&3. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO4	3	3	2	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	3	2	2	1
Justification for correlation	The students apply the	The students evaluate	The students find the	The students demonstrate problem-	The students apply the Files concepts to

	knowledge of Files	given problem statements and identify solutions	solutions for the given task.	solving skills as a team.	solve problems
--	--------------------	---	-------------------------------	---------------------------	----------------

• **Images / Screenshot of the practice:**

L. Sai Vishnu
S. Balaji

Crossword Puzzle- Input Output

Across

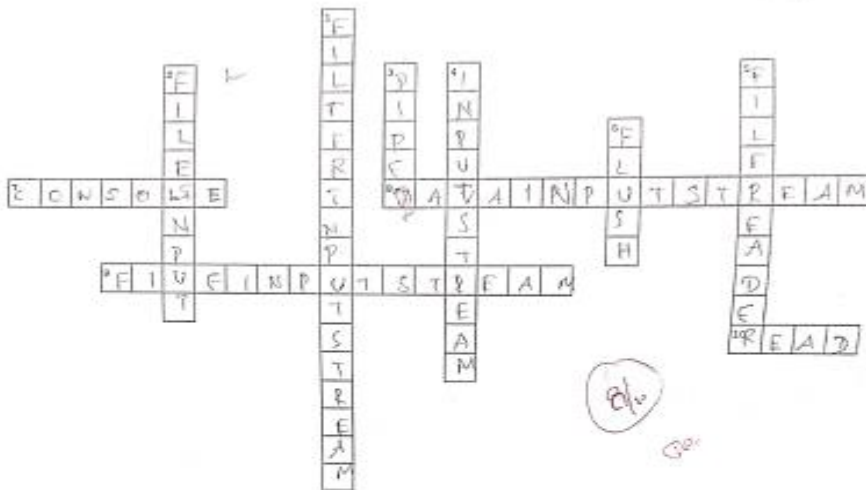
7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionally for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

Crossword Puzzle- Input Output

Podnappiyan C
Mangala G



Across

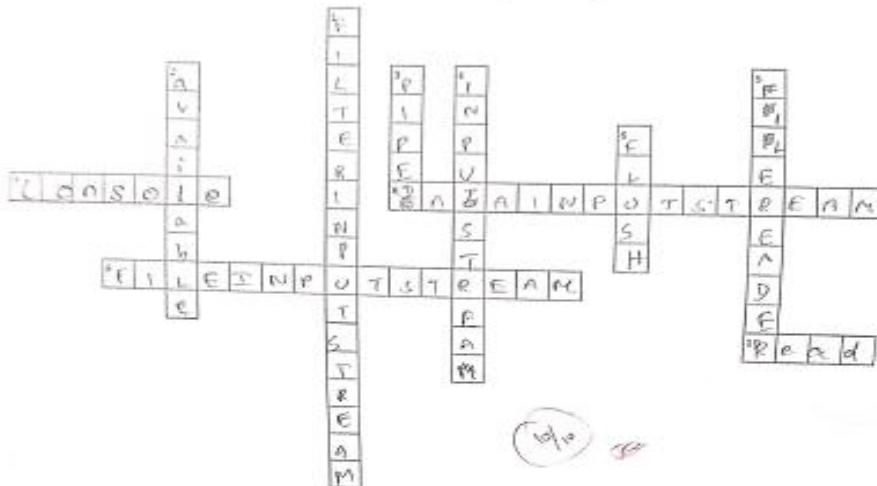
7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionally for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

95 36 1 3 1 0 60 66
95 36 1 3 1 0 60 67

Crossword Puzzle- Input Output



Across

7. class is used to read password.
8. class provides the capacity to read primitive data types from an input stream.
9. is used for reading byte-oriented data from a file.
10. method of InputStream is used to read integer representation of next available byte input

Down

1. The class DataInputStream extends and implements the interface DataInput.
2. method returns the estimated number of bytes that can be read from the input stream.
3. streams provide functionally for threads to communicate and exchange data between them.
4. abstract class is the superclass of all classes representing an input stream of bytes.
5. class is used for reading streams of characters
6. method force writes whenever the data accumulates in the output stream.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Some of the students find very easy to identify the classes used for input and output.
- Some of them take time to complete the puzzle need more clarification on the various classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The student level of understanding on various input output stream classes improved. They can easily identify the place where they used which type of stream classes.

- ❖ ***Challenges faced in implementation:***

- Some students didn't understand the two type of IO stream classes struggle to find answer to the puzzle.
- Some of them feel difficult to identify the file classes and its method such as reading and writing.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/48.GE8151_Crossword.pdf